Statement of Work

For: Texas Instruments

Project: TI TVM Abstraction Layer

SMU Computer Science Senior Design Team
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Document History
February 9, 2026: Initial Draft Created
February 16, 2026: Updated document cover page, purpose of document, scope, modified services
____: Final document

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Purpose of Document

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The purpose of this document is to present the Statement of Work (SOW) which outlines the SMU Computer Science Senior Design Team's roles, tasks, dependencies, and deliverables at a high level for Texas Instruments. Texas Instruments will enable the SMU Computer Science Senior Design Team to design and eventually implement an open-source abstraction layer for the AM69A (TDA4x-class) platform that enables execution of quantized 8 bit and 16 bit ONNX models across processors, including ARM Cortex-A72 cores and C7x DSP cores with MMA acceleration where available.

The SMU Computer Science Senior Design Team is not obligated to provide services described in this SOW unless an order for the service, incorporating the terms of an agreed SOW, has been placed by the customer under a signed governing agreement in place between Texas Instruments and the SMU Computer Science Senior Design Team and accepted by the SMU Computer Science Senior Design Team. The SMU Computer Science Senior Design Team's performance of the Services described herewith is subject to the assumptions, exclusions and other conditions identified in this document. In the event of a conflict between the terms of the Agreement and this SOW, the terms of this SOW shall prevail with respect to the subject matter contained herein.

Scope of Project

The SMU Computer Science Senior Design Team will work with Texas Instruments to deliver an open-source TVM abstraction layer for the AM69A processor that enables efficient deployment of AI/ML models on TI's embedded hardware. The system will provide a unified interface with Python and C++ bindings, allowing developers to leverage the AM69A's specialized computer resources (C7x DSP cores with MMA and ARM Cortex-A72 cores) without requiring low-level hardware programming expertise. The key functionalities include:

1. **Core TVM Integration Layer**

1.1 System Architecture Design

- o Define the high-level system architecture for the TVM integration layer.
- o Document computer routing strategy across:
    - ARM Cortex-A72 cores

- C7x DSP cores
- MMA accelerator (where available)

- Produce architecture & diagrams and design documentation.

1.2 TVM Integration & Runtime Implementation

- Develop a unified Intermediate Representation (IR) interface for the AM69A processor.
- Create a minimal portable runtime optimized for embedded deployment.
- Implement connection interfaces to C7x DSP cores and ARM Cortex-A72 CPU cores.

2. **Language Bindings & Developer Interface**
    - Provide Python bindings for high-level model deployment and execution.
        - Optional: Provide C++ bindings for performance-critical applications.
    - Create a simple, intuitive API that abstracts hardware complexity from end users.

3. **Building System**
    - Integrate as a standard Make-based project compatible with TI Processor SDK.
    - Ensure compatibility with TI’s Top Level Makefile approach described in TI's SDK documentation.
    - Configuration management via .env files (no hardcoded environment variables at runtime).

4. **Testing & Validation**
    - Validate inference execution on:
        - ARM Cortex-A72 cores
        - C7x DSP cores
        - MMA accelerator (if supported)
    - Demonstrate successful execution of at least one quantized ONNX model.
    - Perform functional validation testing to confirm correct inference behavior.

5. **Documentation & Knowledge**

    Provide comprehensive Markdown documentation including:
    - Build instructions and system requirements
    - User's guide with code examples and API reference
    - Architecture overview and design decisions
    - Deployment guide for getting models running on hardware

Out of Scope

Anything that was not mentioned in Section 3 is considered out of scope.

Activities outside the scope of this project include, but are not limited to, the following:

- **Custom DSP Kernel Development:** Writing new low-level DSP kernels or optimized assembly code for the C7x cores or MMA accelerator.
- **ONNX Interpreter Implementation:** Developing a custom ONNX parser, runtime interpreter, or framework independent of TVM.
- **Machine Learning Model Training:** Designing, training, or fine-tuning machine learning models (project focuses on deployment infrastructure only).
- **Hardware Design or Modification:** Any changes to the AM69A hardware platform, physical board, chip design, or electrical characteristics.
- **Graphical User Interface:** No web dashboard, desktop application, or visual debugging tools.
- **Integration with TI Systems:** Connection to internal TI tools, databases, or proprietary infrastructure beyond the publicly available SDK resources.
- **Support for Non-TI Hardware:** Porting to processors outside the TDA4/AM69A family.
- **Long-Term Maintenance and Commercial Support:** Ongoing maintenance, patch management, feature extensions, or post-project technical support beyond the academic semester timeline.

Services Rendered

The services defined in this SOW attend to the fundamental functional areas where the SMU Computer Science Senior Design Team can add value and improve upon Texas Instruments' TVM. The solution will be delivered using a hybrid Agile and Waterfall Software Delivery Life Cycle (SDLC). The key service categories that will be implemented are as follow:

Service Category: Program Management

Object and Value:

The SMU Computer Science Senior Design Team provides program management, which will take care of all the activities assigned to them and agreed upon by Texas Instruments in order to make sure that everything is done with quality and within time and budget. SMU Computer Science Senior Design Team's Program Manager is responsible for supporting the Texas Instruments' TVM service for the scope of this SOW.

As requirements identify the Features and User Stories, a Sprint Development schedule will be created with the understanding that it can change as required by project priorities shift.

The SMU Team will engage in daily meetings starting at the beginning of January and will conclude sessions on April 28th. A final Senior Design Expo to showcase projects will be held on April 30th, 2026.

Key Deliverables:

- Tracking project milestones and deliverables from both SMU Computer Science Senior Design Team and TI TVM. Providing plans for issues related to schedule, cost, and risk management.
- Schedules defining tasks, durations, and deliverable milestone dates.
- Bi-weekly status reports that provide high-level summary on this project.

- Weekly progress meetings.
- High-level system architecture and component interaction diagrams.
- Design tradeoff analysis.
- Deployment and integration plan with TI Processor SDK top-level Makefile framework.

Service Category: Software Engineering

Objective and Value:

The SMU Team will provide an end-to-end software engineering lifecycle solution starting with requirements gathering, software design, software development & testing, deployment, and transition to Texas Instrument's TVM.

Requirements Engineering defines the technical characteristics of the solution. SMU Computer Science Senior Design Team will provide a technical deep dive session with Texas Instrument's TVM to present an overview of the functional and non-functional requirements needed to meet the project scope. Following the deep dive session, weekly meetings will be held to finalize the requirements document. Initial wireframes will be created and modified using 3 iterations before finalizing. Requirements are broken down into Features, and User Stories. The User Interface (UI) mock-ups will be completed in this phase. We will refer to them as wire frames.

Requirements output will lead to features & user stories spread across Sprints for Implementation.

Architectural Design defines the technical characteristics of the Solution. SMU Computer Science Senior Design Team will provide a technical deep dive session with TI TVM to present an overview of the design, explain the services and user interface. Following the deep dive session, weekly meetings will be held to guide TI TVM on the complete design and reach agreement. SMU Computer Science Senior Design Team will utilize its practices and experience to provide the design documentation. The architecture should document with hardware, software, and connectivity details to ensure requirements can be delivered. The output is a context or high-level diagram. If this already exists within the company, no additional work is required.

Implementation and Testing is the code development, unit testing, and system integration for the project. Implementation will be done using Agile Development with 15-day deployments to either production or testing environment. Defect fixing and coding continue throughout the project.

Service Category: Transition

Objectives and Value:

The objective of this service is to provide a transition information and handover to Texas Instruments' TVM. This information will be used to help the Texas Instruments' TVM transfer from students to internal development organization. The key deliverables are the Software Engineering Document that provides an end-to-end document of the project from Requirements to Deployment. The Transition will also identify future enhancements, code location, and other key items needed for Texas Instruments' TVM to update and maintain the solution.

Texas Instruments Responsibilities

The scope and realizations defined in the Services Section are contingent upon the ensuing customer responsibilities.

- TI TVM arranges to have qualified personnel to attend all meetings essential to adhering to full regulatory compliance required for operations. TI TVM must provide a SPOC.
- All project resources and security access necessary to fulfill Scope of Project will be provided by TI TVM where applicable.
- SMU Computer Science Senior Design Team will be responsible for the operations, services, and performance of the products we provide, but not those that are provided and maintained by 3rd party contractors.
- TI TVM is responsible for all user activity associated with the application after SMU Computer Science Senior Design Team delivery.
- TI TVM provides access to any Cloud Services for implementation as required.
- TI TVM will provide access to development, testing, and production environments where required.

Acceptance Criteria

SMU Computer Science Senior Design Team will provide TI TVM a Demo at the end of each Sprint for acceptance.

Agreement

This SOW and the terms and conditions of the Agreement constitute the entire agreement between Texas Instruments TVM and SMU Computer Science Design Team and supersede all prior oral or written negotiations and agreements regarding the subject matter herein. Any modification or addition to this SOW shall be in writing and signed by authorized representatives of both parties.

IN WITNESS WHEREOF, the parties hereto have caused this SOW to be executed by their duly authorized representatives

Signature not required

Texas Instruments TVM	SMU Computer Science Senior Design Team
Signature 1: ______________________	Signature 1: ______________________
Name: ______________________	Name: ______________________
Title: ______________________	Title: ______________________
Date: ______________________	Date: ______________________